

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

Department of Computer Science & Engineering

End-Semester Examination, Spring 2023-24

B. Tech. 6th Semester

Subject Name: Computer Networks

Subject Code: CS3002

Number of pages: 2

Maximum Marks: 50

Exam Duration: 3 Hours

Note: There are Nine questions. **Q.1** is **compulsory**. Attempt any Seven questions from the **Q.2-Q.9**. All parts of the same question must be attempted in a sequence.

Q.1 (a) Differentiate 1-persistent, p-persistent and Non-persistent CSMA.

(b) Consider that the 4-bit generator (G) is 1001, that the data payload (D) is 10011111 and that $r = 3$. What are the CRC bits (R) associated with the data payload D?

(c) Given an IP address: 201.20.30.40/24. Find the Network ID, Fourth Host ID, Last usable Host ID, and Broadcast Address.

(d) Discuss any two Application layer protocols from HTTP, FTP, SMTP, and DNS.

(e) What do you understand by Bit-stuffing?

201.20.30.4

[3x5=15 Marks]

Q.2 Suppose users share a 2.5 Mbps link. Also suppose each user requires 250 kbps when transmitting, but each user transmits only 10 percent of the time.

(a) When circuit switching is used, how many users can be supported?

(b) For the remainder of this problem, suppose packet switching is used. Find the probability that a given user is transmitting.

(c) Suppose there are 20 users. Find the probability that there are more than 17 users transmitting simultaneously.

[1+2+2=5 Marks]

Q.3 A 20 Kbps satellite link has a propagation delay of 400 ms. The transmitter employs the "Go Back N ARQ" scheme with N set to 10. Assuming that each frame is 100 Bytes long, what is the maximum data rate possible?

$10 * T_t / 2 * PD * bw$

[5 Marks]

Q.4 Calculate RTO after receiving the last ACK. First ACK is received after 20 ms (RTT). The 2nd and 3rd ACK are received at 25ms, and 30 ms, respectively. Use the standard parameter values $\alpha=1/8$ and $\beta=1/4$ for computation of estimated RTT and DevRTT.

[5 Marks]

Q.5 Consider sending an object of size $O = 128$ Kbytes from server to client. Let $S=736$ bytes and $RTT=588.8$ msec. Suppose the transport protocol uses static windows with window size W . For a transmission rate of 20 Kbps, determine the minimum possible latency. Determine the minimum window size that achieves this latency.

[2+3=5 Marks]

Q.6 Consider a network with the CIDR address 194.100.128.0/19. An organization wants 6 subnets, and each subnet should have at least 22 hosts. What subnet mask should the organization use to fulfil the above condition? Also write the addresses of each subnet for efficient design of address.

/27

[5 Marks]

✓ **Q.7** Suppose that host A is connected to a router R1; R1 is connected to another router, R2; and R2 is connected to host B. Suppose that a TCP message that contains 980 bytes of data and 20 bytes of TCP header is passed to the IP code at host A for delivery to B. Show the Total length, Identification, DF, MF, and Fragment offset fields of the IP header in each packet transmitted over the three links. Assume that links A-R1, R1-R2, and R2-B have the MTU of 1500 Bytes, 640 Bytes, and 680 Bytes, respectively.

[5 Marks]

✓ **Q.8** Consider the network scenario of three nodes A, B and C connected linearly as given in Fig. 1 and each link has link cost of 1 unit. A is connected through an internet router R. Show the delay in routing tables at A, B, and C with respect to router R respectively in the following cases:

- (a) If an internet router is down initially.
- (b) If an internet router is up then showing the routing tables up to 3 exchanges.
- (c) If an internet router is down again then show the routing tables up to the next 3 exchanges.

[1+2+2=5 Marks]

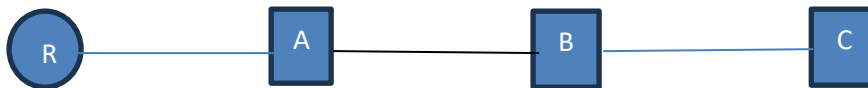


Fig. 1

✓ **Q.9** Two nodes A and B using CSMA/CD medium access protocol are attached to opposite ends of the Ethernet Bus Network. The propagation delay is $22.5\mu s$. Both nodes attempt to transmit frame at time $t=0$. Frames collide and after collision, A gets $K=0$ and B gets $K=1$. Consider Jamming signal and inter-frame space (IFS) or idle slot. At what time does B schedule its retransmission? Also determine the time at which A's frame of 512 bits will be delivered at B on a link of 10 Mbps. Will there be a collision during retransmission? Justify your answer.

[2+2+1=5 Marks]